

Lu-Ching Wang

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Research Interests

Rescue Robots, Vision-Language Navigation, Quadruped Robots, Reinforcement Learning, Visual Servoing, Control Theory.

Education

- University of Bonn (Uni Bonn)** **Bonn, Germany**
 PhD, Department of Computer Science Nov. 2026 – Present
[Humanoid Robots Lab](#), Advisor: Dr. [Maren Bennewitz](#)
- National Taiwan University (NTU)** **Taipei, Taiwan**
 Master of Science, Department of Electrical Engineering (GPA: 4.1/4.3) Sep. 2022 - Aug. 2024
[Networked Control Systems Lab \(NCS Lab\)](#), Advisor: Dr. [Feng-Li Lian](#)
 Master's thesis: [Hybrid Visual Servo Manipulation for Isotropic Target Using Individual Spatio-Temporal Image Feedback](#)
- National Yang Ming Chiao Tung University (NYCU)** **Hsinchu, Taiwan**
 Bachelor of Science, Department of Mechanical Engineering (GPA: 3.9/4.3) Sep. 2016 - Jun. 2021
- Technische Universität Berlin (TU Berlin)** **Berlin, Germany**
 Exchange Student Program, Department of Mechanical Engineering Sep. 2019 - Oct. 2020

Research & Working Experience

- Inventec Corporation** **Taipei, Taiwan**
 Robotics Research Engineer, [AI Center, Group of Robotics](#), Advisor: Dr. [Wei-Chao Chen](#) Aug. 2024 - Present
 - Benchmarking Vision-Language Navigation with 3D Gaussian Reconstruction for Deployment**
 - Evaluated the 3DGUT GS pipeline using reconstruction metrics; collected custom datasets and performed GLOMAP.
 - Conducted real-world VLN infrastructure deployment and experimental evaluations; results published at [1].
 - Feasibility-Guided Planning over Multi-Specialized Locomotion Policies**
 - Developed a learning-based planning system that selects the optimal path and locomotion policy for traversing diverse terrains.
 - Maintained interpretability while adding new policies; achieved a success rate of 99.8% in simulations and 70% in experiments.
 - Executed real-world experiments involving mapping, navigation, and policy switching; results published at [2].
 - Quadruped Robots SDK & Sim-to-Real Deployment**
 - Built a quadruped robot interface for customized actuators and sensors through LCM communication.
 - Trained locomotion policies in IsaacGym and IsaacLab; deployed in C++ with parameters tuning (action scales, motor gains).
 - Quadruped Robot Challenges (QRC)**
 - Team leader and operator of a 4-members team. Deployed the planner based on *Feasibility-Guided Planning* research.
 - Integrated ROS2 for navigation, locomotion, manipulator teleoperation, and image detection over extreme terrains.
- National Taiwan University (NTU)** **Taipei, Taiwan**
 Master's Student, [Networked Control Systems Lab \(NCS Lab\)](#), Advisor: Dr. [Feng-Li Lian](#) Sep. 2022 - Aug. 2024
 - Mobile Armed Robot for Tomato Harvesting**
 - Built a tomato harvesting wheeled-mobile robot for greenhouse scenario, achieving a 68.4% harvesting success rate.
 - Reduced the harvesting time from 21.2 to 6.26 seconds under a 60kg payload; results published at [2] and [3].
 - Teaching Assistant of course "Control System" (EE 3024)**
- Academia Sinica (National Academic Institution)** **Taipei, Taiwan**
 Research Assistant, [Research Center for Information Technology Innovation](#), Advisor: Dr. [Yennun Huang](#) Jan. 2022 - Aug. 2022

Publications

- Yu-Lun Chou*, **Lu-Ching Wang***, Ying-Sheng Luo*, Yu-Tang Liu, and Jeng-Lin Lee, "*VL-N3RD-Bench: Benchmarking Vision-Language Navigation with 3D Gaussian Splatting Reconstruction for Deployment*", in 2026 ICRA Workshop: From Data to Decisions: VLA Pipelines for Real Robots, June 2026. (*denotes equal contribution) ([link](#))
- Ying-Sheng Luo*, **Lu-Ching Wang***, Hanjaya Mandala*, Yu-Lun Chou*, Guilherme Christmann*, Yu-Chung Chen, Yung-Shun Chan, Chun-Yi Lee, and Wei-Chao Chen, "*Feasibility-Guided Planning over Multi-Specialized Locomotion Policies*", in 2026 IEEE Int. Conf. Robot. Autom. ICRA, June 2026. (*denotes equal contribution) ([link](#))
- Lu-Ching Wang**, Yu-Cheng Chang, and Feng-Li Lian, "*A Perception and Alignment Framework for Fruit Harvesting Using Spherical Object Modeling and Hybrid Visual Servoing*", in 2025 IEEE/ASME (AIM) Int. Conf. on Advanced Intelligent Mechatronics, July 2025. ([link](#))
- Lu-Ching Wang**, Yen-Cheng Chu, Yennun Huang, and Feng-Li Lian, "*Enhancement on Target-Gripper Alignment: A Tomato Harvesting Robot with Dual-Camera Image-Based Visual Servoing*", in 2024 IEEE Int. Conf. Robot. Autom. ICRA, May 2024. ([link](#))

Awards & Competitions

- 2025 ICRA Quadruped Robot Challenge (QRC) – Participation Certificate, Autonomous** ([link](#))
- 2024 IROS Quadruped Robot Challenge (QRC) – 3rd Placement, Teleoperation** ([link](#))
- 2024 ARIS Best Student Paper Award – 1st Place**, 2024 Int. Conf. on Advanced Robotics and Intelligent Systems (ARIS) ([link](#))
- 2024 Master Thesis Award – Distinction Award** from Robotics Society of Taiwan (RST) ([link](#))

Technical Skills

- Software:** Python (NumPy, Pytorch, OpenCV, Matplotlib, pip), C++ (PyTorch/ATen, STL), MATLAB (Simulink), CMake, Git, Blender
- Robotics:** Newton, IsaacSim, IsaacLab, IsaacGym, ROS2 (Fast-LIO, RTAB-Map, Nav2, MoveIt, URDF)
- Hardware:** RTOS, multi-threading, RS-485, Serial, Lightweight communications and marshalling (LCM), UDP